

Performance Benchmarking of Embedded Battery State Estimation: A Measurement-Only Comparison of Direct-Measurement, Hybrid, and Data-Driven Estimators

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Abstract

This paper presents a measurement-only performance benchmark for embedded battery state estimation under realistic Battery Management System (BMS) constraints. The study isolates estimator performance rather than hardware-resource or implementation-effort evaluation and compares four embedded-ready estimators on the same Lithium Iron Phosphate (LFP) full-cell trajectory under a shared embedded constraint: a direct-measurement model (DM), a hybrid direct-measurement model with learned State-of-Health support (HDM), a hybrid equivalent-circuit model with voltage feedback (HECM), and a data-driven neural model with GRU-based SOC estimation and LSTM-based online SOH context (DD). The benchmark perturbs only measurable inputs and timing information, including ADC quantization, current bias, additive current, voltage, and temperature noise, missing samples, irregular sampling, burst dropout, voltage spikes, and initial-state mismatch, so that all disturbances remain physically interpretable and reproducible. The comparison reveals no universal winner across deployment-relevant dimensions. Under clean conditions, HDM provides the lowest baseline error with an MAE of 0.0024, followed by HECM with 0.0090,

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DD with 0.0100, and DM with 0.0311. Under disturbed measurements, HECM is the strongest overall robustness choice, especially in the most discriminative current-bias and voltage-noise scenarios. DD is the most stable estimator under missing samples and recovers fastest after initialization mismatch, while DM shows the expected drift sensitivity and HDM trades nominal accuracy against reduced disturbance tolerance. The benchmark therefore exposes complementary failure modes of direct-measurement, hybrid physics-based, and data-driven estimators and supports decision-oriented model selection for embedded BMS deployment.

Keywords: Battery management system, State of charge, State of health, Performance benchmark, Direct measurement, Equivalent circuit model, Data-driven estimation, Embedded artificial intelligence

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CRedit authorship contribution statement

Florian Rzepka: Conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing–original draft preparation, writing–review and editing, visualization.

Qinnan Chen: Writing–review and editing.

Julia Kowal: Resources, supervision, project administration, funding acquisition, writing–review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used ChatGPT, DeepL, and Grammarly in order to improve the linguistic quality and style of the manuscript. After using these tools, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Data availability

The underlying laboratory dataset, benchmark scripts, model runners, and paper-facing result exports will be made available on reasonable request.